

BLOOD DONATION SYSTEM

Software Engineering (SE) Project

Project Title : Blood Donation Management System

Submitted By:

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Semester: 3C

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Subject: Software Engineering

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“A system designed to efficiently manage blood donors, recipients, blood requests, and blood availability to save human lives.

• Blood Donation Management System

Introduction:

The **Blood Donation Management System** is a software project designed to manage and organize the process of blood donation efficiently. It connects **donors, recipients, and blood banks/hospitals** through an online platform. The main goal is to make it easier to **find donors quickly** in times of emergency and **maintain accurate blood stock records** in hospitals and blood banks.

This system helps in:

- Registering blood donors and recipients.
- Managing blood group data and availability.
- Tracking donation history and requests.
- Ensuring timely communication between donors and hospitals.

The system reduces manual work, improves accuracy, and speeds up the blood donation process, ultimately saving lives.

• SDLC Model Used: Waterfall Model

The **Waterfall Model** is chosen for this project because each phase is completed before the next begins. Requirements, design, and development steps are well defined. Our Blood Donation System has clear and stable requirements. Progress is tracked step-by-step through each stage. Ensures quality and correctness before moving forward.

- **Functional Requirements**

Functional requirements describe **what the system should do** , the main features and functions.

1. **User Registration and Login:**

- Donors, recipients, and admins can create accounts and log in.

2. **Donor Management:**

- Add, update, or delete donor information (name, age, blood group, contact, city).

3. **Blood Request Management:**

- Recipients can request blood by selecting blood group and location.

4. **Blood Stock Management:**

- Admin can manage available blood units for each group.

5. **Search Function:**

- Users can search for donors by blood group, city, or availability.

6. **Notification System:**

- Send SMS or email alerts to donors for donation camps or urgent needs.

7. **Donation History:**

- Track each donor's past donations and eligibility for the next donation.

8. **Admin Dashboard:**

- Manage users, view reports, and monitor system activity.

- **Non-Functional Requirements**

These define **how the system should perform** — focusing on quality, speed, and reliability.

1. **Performance:**

- The system should respond within 2–3 seconds for any user request.

2. **Reliability:**

- System should be available 24/7 with minimum downtime.

3. **Security:**

- Data like user details and passwords must be encrypted and protected.

4. **Usability:**

- Interface should be simple and easy to navigate for all users.

5. **Scalability:**

The system should handle increasing numbers of users without slowing down.

6. **Availability:**

- Should work continuously without interruptions, especially in emergencies.

7. **Maintainability:**

- Easy to update features or fix bugs in the future.

8. **Compatibility:**

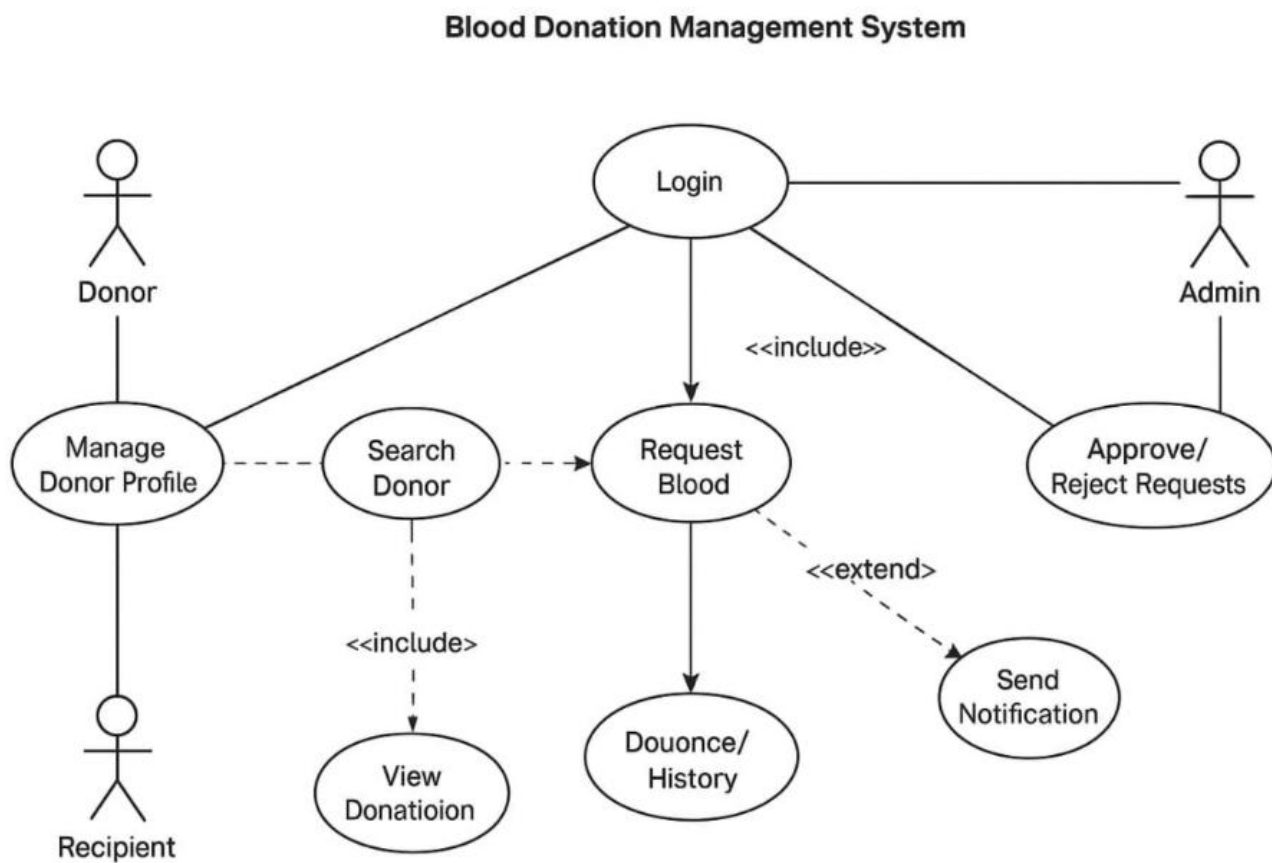
- Should run on multiple browsers like Chrome or other mobile devices.

Domain Requirements

1. **Donor Eligibility Rules**

- Donors must be healthy, 18–60 years old, and meet basic health conditions.
- They can donate blood only after every 3 months.
- 2. Blood Group Handling**
 - System must support all blood groups (A, B, AB, O with + / –).
 - Blood must match correctly with the recipient.
- 3. Blood Stock Management**
 - System must record available blood units, their type, and expiry date.
 - Expired or unsafe blood must not be used.
- 4. Donation Process Rules**
 - Real steps must be followed: registration → checkup → collection → testing → storage → issue.
- 5. Blood Safety Requirements**
 - All blood must be tested for infections like HIV, Hepatitis, etc.
 - Unsafe blood must be removed immediately.
- 6. Emergency Request Handling**
 - System must quickly show available donors or blood units during urgent requests.
- 7. Hospital/Blood Bank Rules**
 - Only verified hospitals and blood banks can request or update blood stock.
- 8. Notifications**
 - Donors must receive alerts for emergencies or when they are eligible to donate again.
- 9. Privacy Protection**
 - Donor and patient personal information must be kept secure and private.
- 10. Record Keeping**
 - System must maintain complete records of each blood unit: donor, test results, date, and usage.

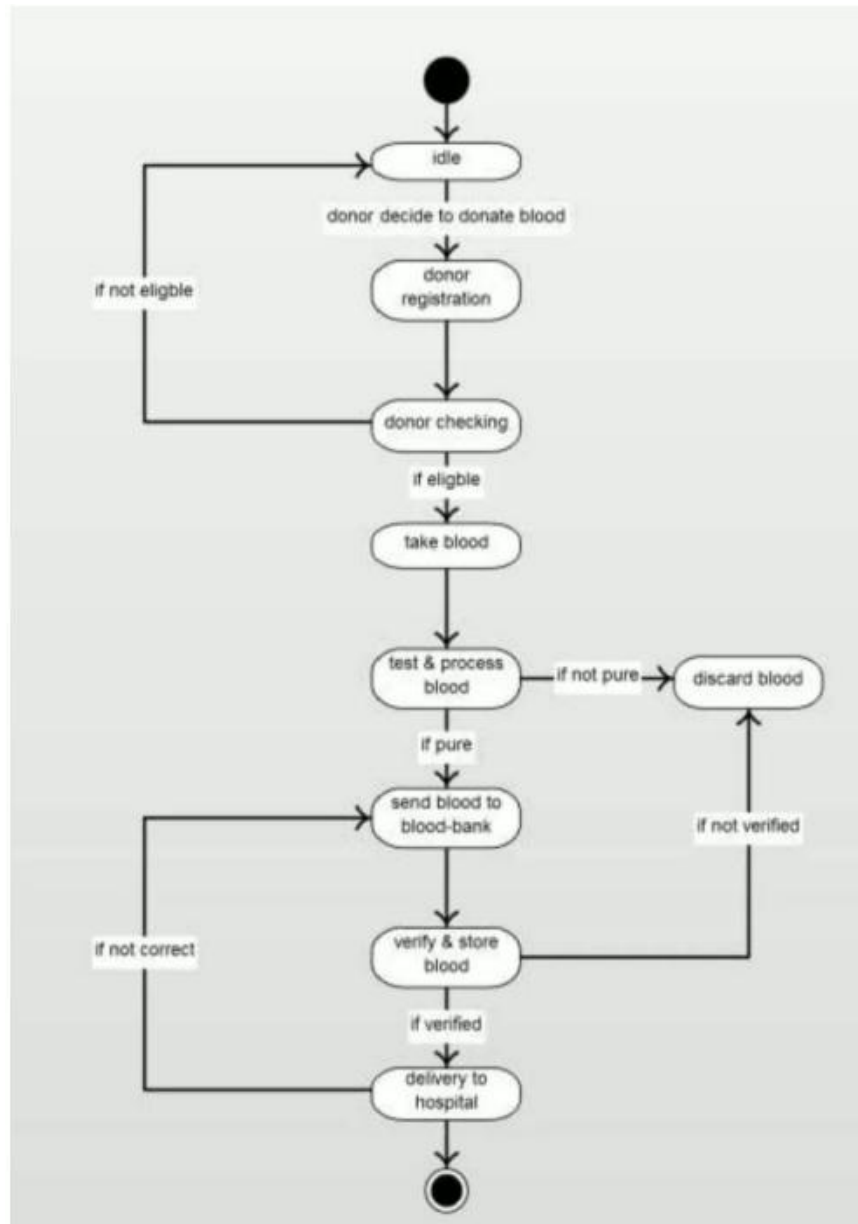
■ UseCase Diagram For Blood Donation System



■ USE CASE DIAGRAM EXPLANATION

Use Case ID	Use Case Name	Actor	Description	Precondition	Postcondition
UC-01	Manage Profile	Donor	Create or update personal details	User logged in	Profile updated
UC-02	Search Donor	Donor / Recipient	Search donors in the system	User logged in	Donor list displayed
UC-03	Request Blood	Donor / Recipient	Submit a blood request	User logged in	Request submitted
UC-04	View Donation History	Donor	View donation records	Donor logged in	History displayed
UC-05	View Donation Status	Recipient	View blood availability/status	Recipient logged in	Status displayed
UC-06	Login	Admin	Access system features	Valid credentials	Dashboard accessed
UC-07	Approve / Reject Request	Admin	Manage blood requests	Admin logged in	Request status updated
UC-08	Send Notification	Admin	Send alerts and updates	Admin logged in	Notification sent

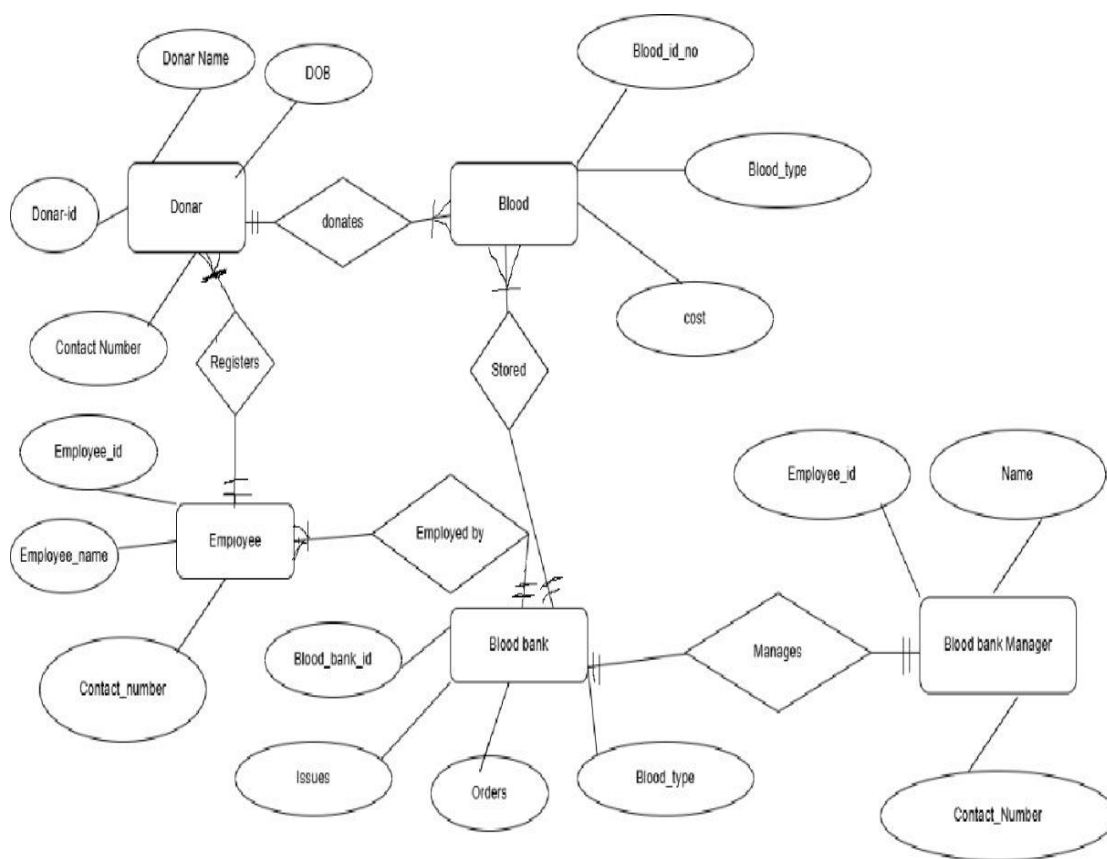
■ Activity Diagram For Blood Donation System



▪ ACTIVITY DIAGRAM EXPLANATION

Phase	Step	Activity	Description
Donor Process	1	Start	Donor decides to donate blood
	2	Registration	Donor registers in the system
	3	Eligibility Check	Donor eligibility is verified
	4	Blood Collection	Blood is collected from donor
Blood Processing	5	Blood Testing	Blood is tested for safety
	6	Blood Storage	Blood verified and stored in blood bank
Distribution	7	Blood Delivery	Blood is delivered to hospital
Completion	8	End	Process completes

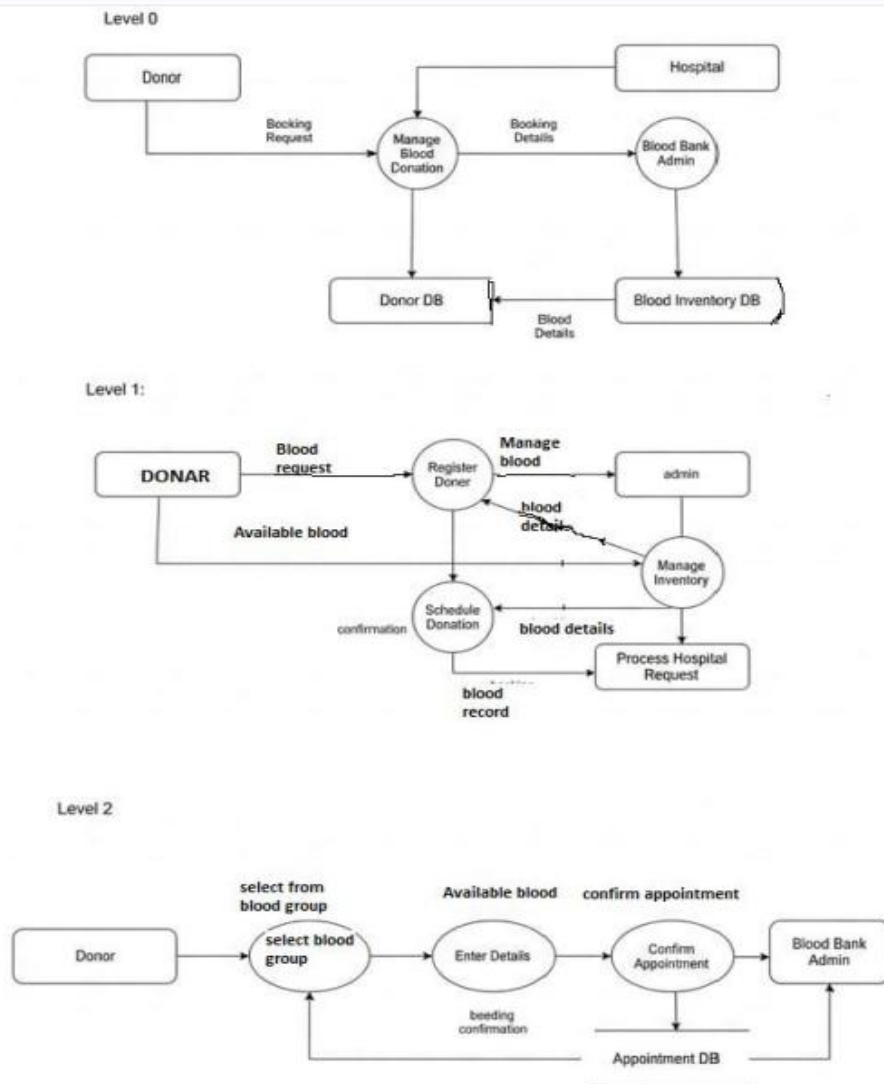
▪ ERD For Blood Donation System



▪ Relationship–Cardinality Table

Relationship	Entities Involved	Cardinality	Explanation
Donates	Donor — Blood	1 : M	One donor can donate many blood units; each blood unit comes from one donor.
Requests	Donor — Employee	M : M	Donors can request help from multiple employees; employees can help many donors.
Employed by	Employee — Blood Bank	M : 1	Many employees work in one blood bank; each employee works at only one bank.
Stored	Blood — Blood Bank	M : 1	Many blood units are stored in one blood bank; each unit is stored in one bank.
Manages	Manager — Blood Bank	1 : 1	Each bank is managed by one manager; each manager manages one bank.

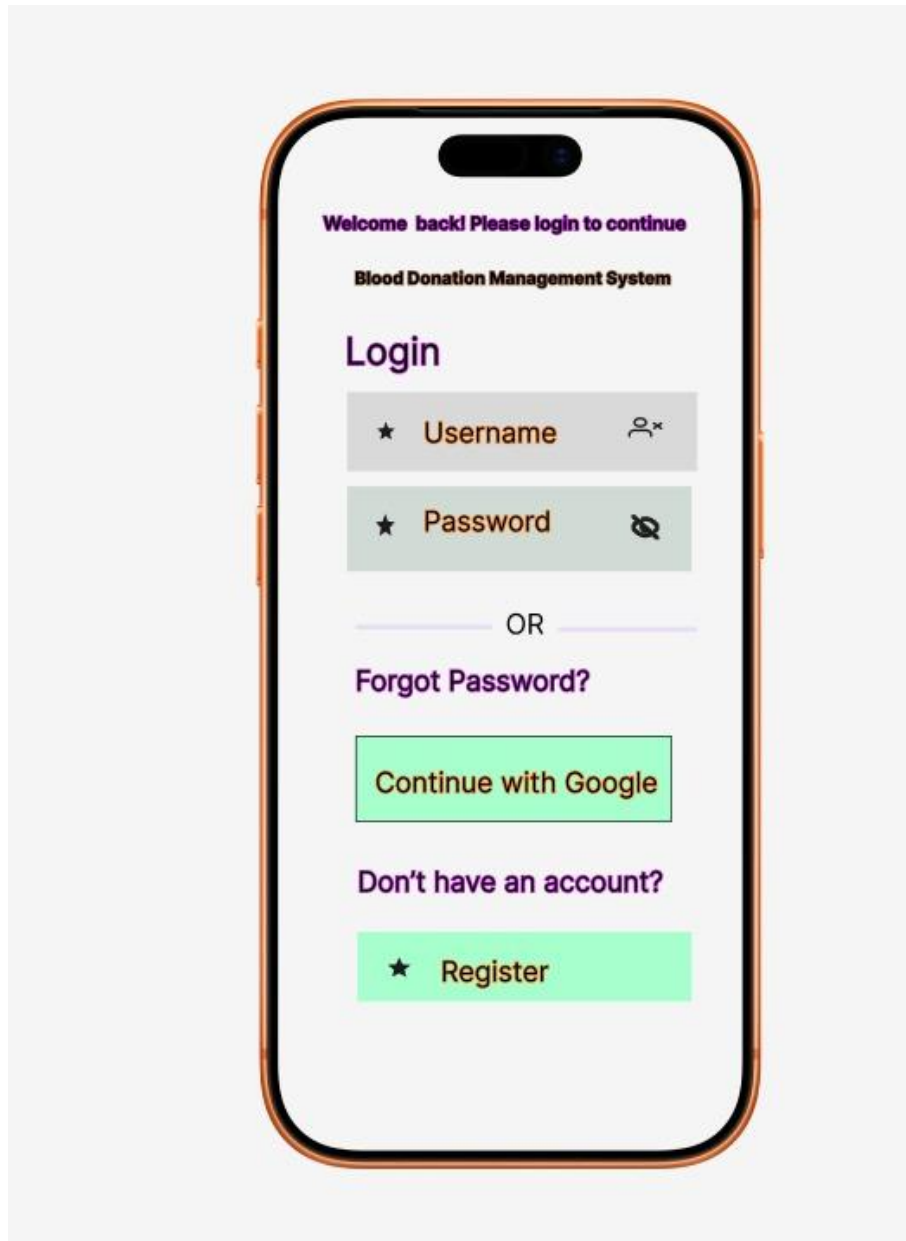
■ DFD For Blood Donation System



▪ **DFD – Blood Donation Management System**

Level	Process / Entity	Description
Level 0	Donor, Hospital	Entities interacting with the system
	Manage Blood Donation	High-level process managing donation flow
	Donor DB, Blood Inventory DB	Databases managed by Blood Bank Admin
Level 1	Register Donor	Handles donor registration and booking requests
	Schedule Donation	Manages donation scheduling and records
	Manage Inventory	Processes hospital blood requests with admin oversight
Level 2	Donor Appointment Booking	Detailed process for donor appointments
	Select Slot → Enter Details → Confirm Appointment	Steps for booking an appointment
	Appointment DB	Updates and stores appointment details

Explanation Mockup screen 1:

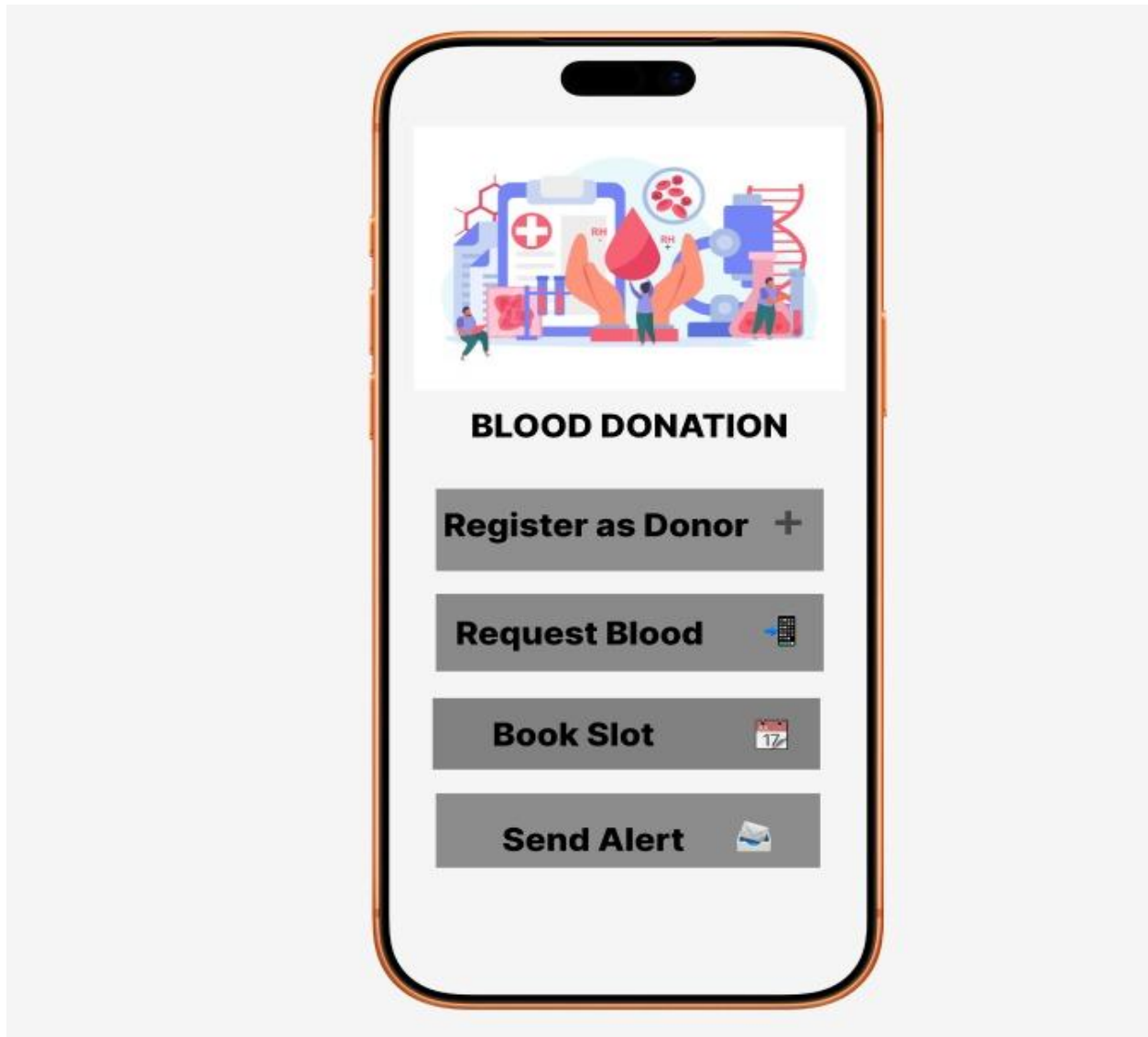


Login/Authentication Screen

Purpose: Provides secure access to the system for registered users.

This screen allows registered users to securely access the system. It includes fields for username and password, a *Forgot Password* option for recovery, Google sign-in for quick access, and a *Register* link for new users. The design is simple and user-friendly to ensure easy and secure login.

Explanation Mockup screen 2:

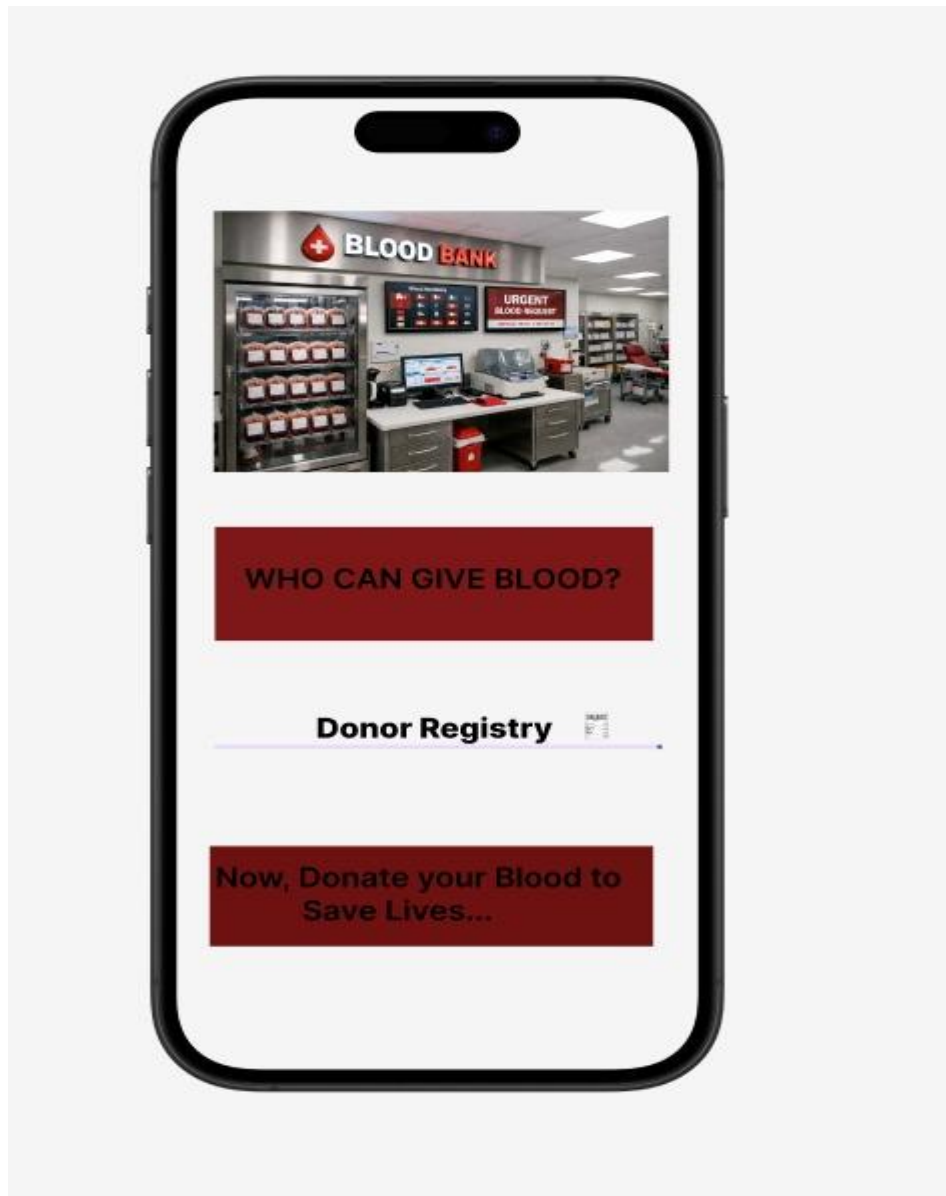


Home / Navigation Screen

Purpose: This is the main interface where users can quickly access the core features of the Blood Donation Management System

.This screen serves as the main interface of the system, allowing users to easily access core features of the Blood Donation Management System. It provides options to register as a donor, request blood, book donation slots, and send alerts for urgent needs. The layout is clean and intuitive, ensuring smooth navigation and quick access to essential functions.

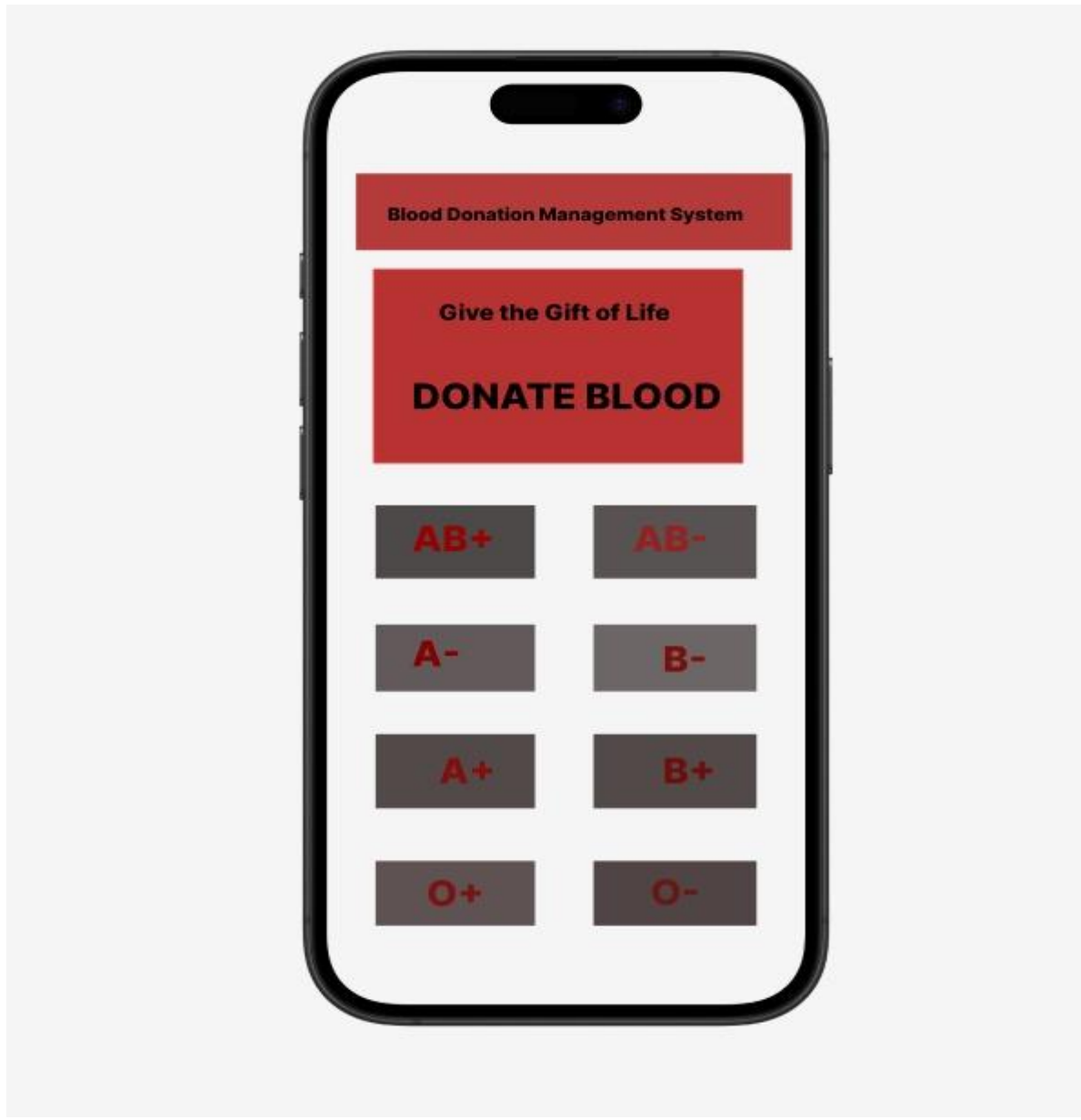
Explanation Mockup screen 3:



Donor Information / Registry Screen

This screen provides information about blood donation eligibility and encourages users to become donors. It displays a blood bank–related image with a highlighted heading, “*Who Can Give Blood?*”, followed by a donor registry section. A motivational message, “*Now, Donate Your Blood to Save Lives*”, is included to inspire users to register and contribute to the blood donation process.

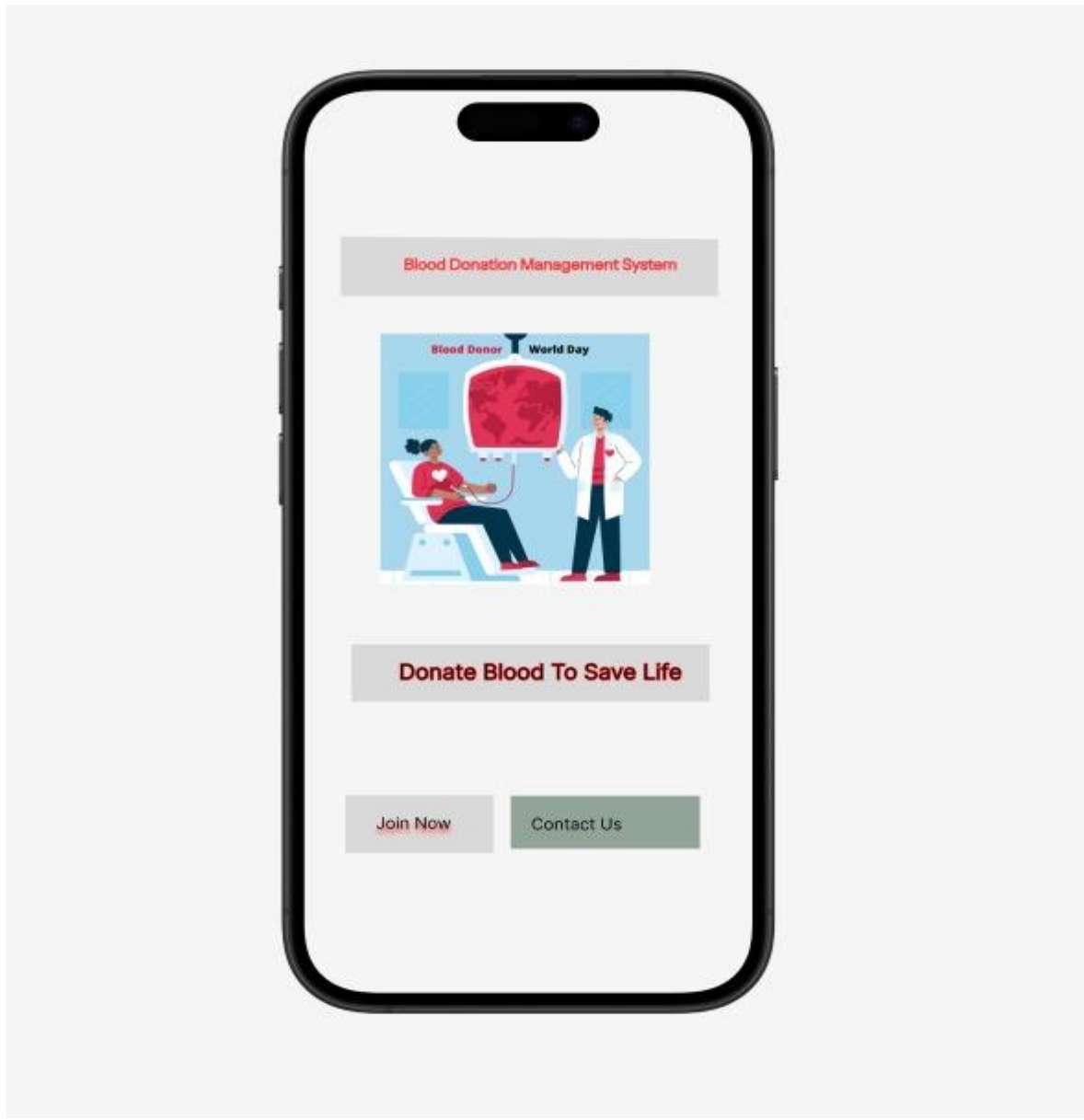
Explanation Mockup screen 4:



Blood Group Selection / Donation Screen

This screen allows users to select their blood group as part of the blood donation process. It displays a motivational message, “*Give the Gift of Life – Donate Blood*”, followed by selectable buttons for different blood groups, including A, B, AB, and O with positive and negative types. The selected blood group helps the system proceed with donation registration or blood availability management.

Explanation Mockup screen 5:



Home / Welcome Screen

This screen serves as the landing page of the Blood Donation Management System. It displays the system title at the top along with an illustration related to blood donation and healthcare. A motivational message, *“Donate Blood to Save Life”*, is shown to encourage participation. The screen includes action buttons such as *Join Now* for donor registration and *Contact Us* for user support or inquiries.

Blood Donation Management System - Test Cases

TC-01 - Register new donor

Input: Valid donor details (name, age, blood group, contact, city)

Expected Result: Donor account created successfully

TC-02 - Register without required fields

Input: Empty name or email

Expected Result: Error message displayed

TC-03 - Valid login

Input: Correct username and password

Expected Result: User successfully logged in

TC-04 - Invalid login

Input: Wrong password

Expected Result: Access denied message shown

TC-05 - Add donor

Input: Valid donor information

Expected Result: Donor added successfully

TC-06 - Update donor details

Input: Modified donor data

Expected Result: Donor information updated

TC-07 - Delete donor

Input: Existing donor ID

Expected Result: Donor removed from system

TC-08 - Request blood

Input: Blood group + location

Expected Result: Blood request submitted successfully

TC-09 - Invalid blood request

Input: Missing blood group

Expected Result: Error message displayed

TC-10 - Add blood stock

Input: Blood group and units

Expected Result: Stock updated successfully

TC-11 - Check availability

Input: Blood group

Expected Result: Available stock displayed

TC-12 - Search donor by blood group

Input: A+

Expected Result: Matching donors displayed

TC-13 - Search non-existing donor

Input: Invalid blood group

Expected Result: No records found

TC-14 - Send notification

Input: Emergency blood request

Expected Result: Notification sent successfully

TC-15 - View donation history

Input: Donor ID

Expected Result: Donation history displayed

TC-16 - Age validation check

Input: Age below 18

Expected Result: Donor rejected (not eligible)

TC-17 - Blood compatibility check

Input: Wrong blood group match

Expected Result: Request denied